Chapter 9
The Open Economy, the basics

Outline

The foreign exchange market.

Fluctuations in nominal exchange rates.

Real exchange rates.

Exchange rate policy.

Balance of payments.

The Foreign Exchange Market

To buy goods and services produced in another country we need the money of that country.

Foreign bank notes, coins, and bank deposits are called **foreign currency**.

Trading Currencies

The **foreign exchange market** is the market in which the currency of one country is exchanged for the currency of another.

The Foreign Exchange Market

Nominal Exchange Rates

The price at which one currency exchanges for another is called a **nominal foreign exchange rate**.

A fall in the value of one currency in terms of another currency is called **currency depreciation**.

A rise in value of one currency in terms of another currency is called **currency appreciation**.

The Foreign Exchange Market

Questions about Exchange Rate

How is the exchange rate determined?

Why does the currency appreciate and depreciate?

How does the central bank operate in the foreign exchange market?

How do fluctuations in exchange rate influence the balance of trade and the balance of payments?

The Foreign Exchange Market

Nominal Exchange Rate Is a Price

An exchange rate is the price of one currency in terms of another.

Like all prices, an exchange rate is determined in a market—the **foreign exchange market**.

The Canadian dollar is demanded and supplied by thousands of traders every hour of every day.

With many traders and no restrictions, the foreign exchange market is a *competitive market*.

The Foreign Exchange Market

When people who are holding one currency want to exchange it for Canadian dollars, they demand for Canadian dollars and supply another currency.

Factors that influence the demand for Canadian dollars also influence the supply of foreign currencies.

Factors that influence the demand for another currency also influence the supply of Canadian dollars.

The Foreign Exchange Market

Demand in the Foreign Exchange Market

The quantity demanded of a currency depends on

1. The exchange rate
2. World demand for home country exports
3. Interest rates in home country and other countries
4. The expected future exchange rate

The Foreign Exchange Market

The Law of Demand for Foreign Exchange

The demand for a currency is a *derived demand*.

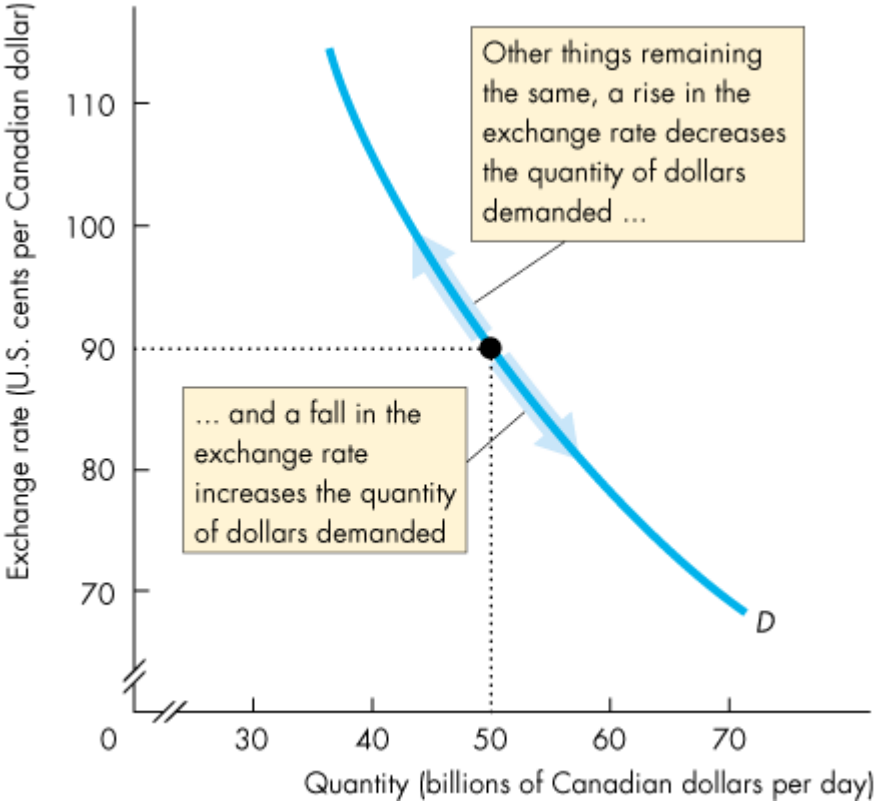
Example. People buy Canadian dollars so that they can buy Canadian-produced goods and services or Canadian assets.

Other things remaining the same, the higher the exchange rate, the smaller is the quantity of Canadian dollars demanded in the foreign exchange market.

The Foreign Exchange Market

The Demand Curve for Canadian Dollars

* Higher exchange rate is an appreciation of Canadian dollar



The Foreign Exchange Market

Supply in the Foreign Exchange Market

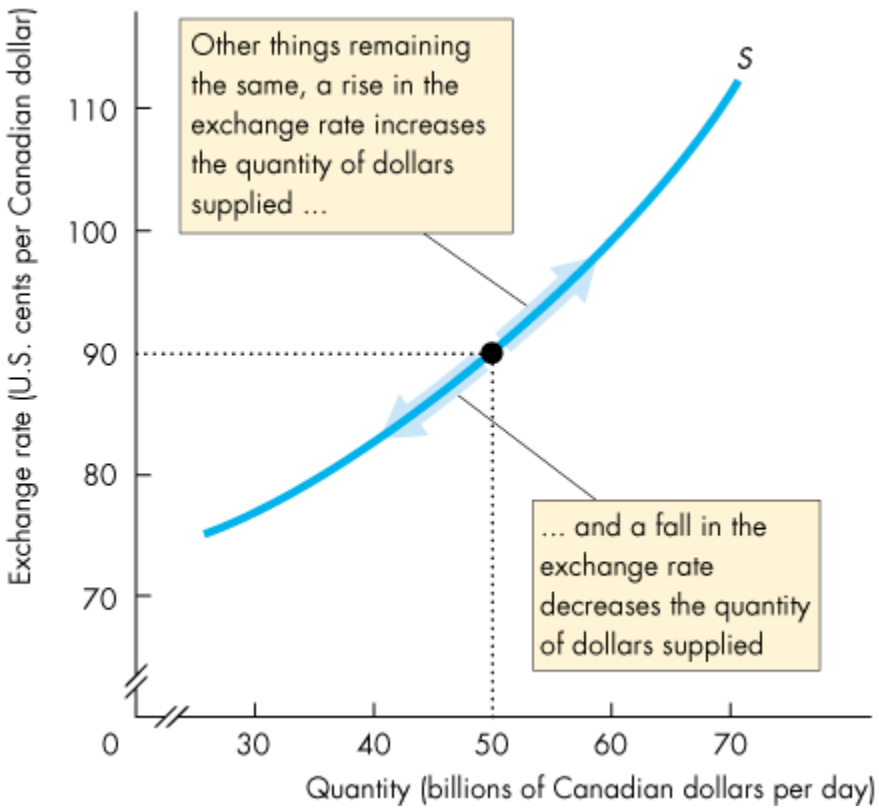
The quantity of a currency supplied in the foreign exchange market depends on many factors but the main ones are

1. The exchange rate
2. Domestic demand for imports
3. Interest rates in domestic economy and other countries
4. The expected future exchange rate

The Foreign Exchange Market

Supply Curve for Canadian dollars

* A higher exchange rate is an appreciation of domestic currency



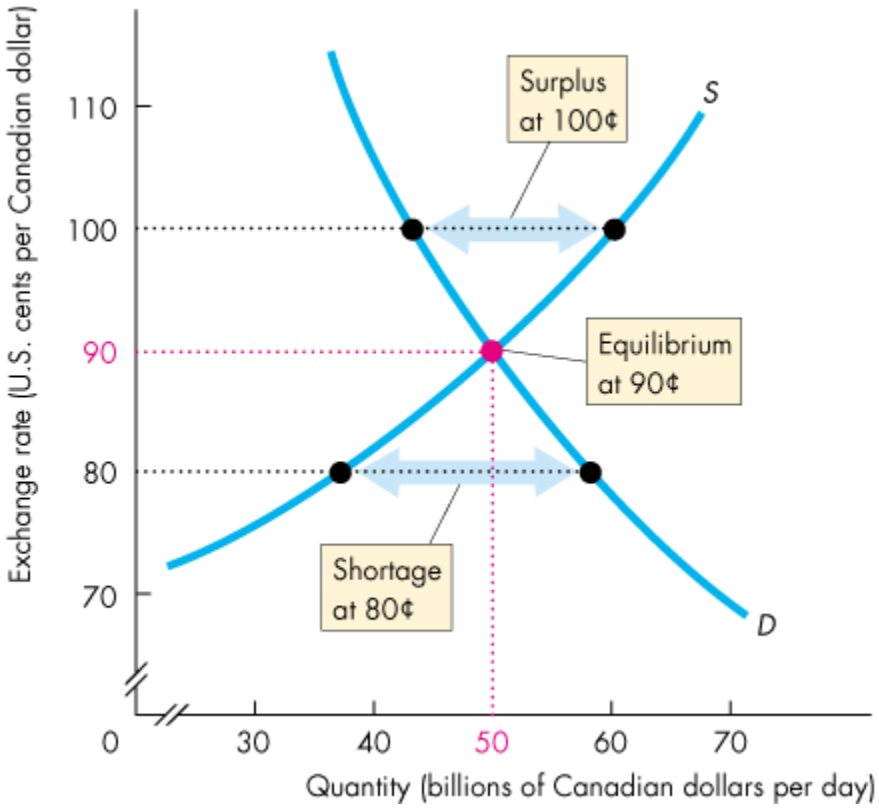
The Foreign Exchange Market

Equilibrium

If nominal exchange rate is too high, a surplus of Canadian dollars drives it down.

If the exchange rate is too low, a shortage of Canadian dollars drives it up.

Market is pulled to equilibrium exchange rate at which there is neither a shortage nor a surplus.



Exchange Rate Fluctuations

Changes in the Demand for a currency

A change in any influencing factors on the quantity of domestic currency, other than the exchange rate, brings a change in the demand for domestic currency.

These influencing factors include

- World demand for domestic exports
- Domestic interest rate relative to the foreign interest rate
- The expected future exchange rate

Nominal Exchange Rate Fluctuations

World Demand for domestic Exports

At a given exchange rate, if world demand for domestic exports increases, the demand for domestic currency increases. Demand curve shifts to the right.

Domestic Interest Rate Relative to the Foreign Interest Rate

Domestic interest rate minus the foreign interest rate is called the **domestic interest rate differential**.

If domestic interest differential rises, the demand for domestic currency increases. Demand curve shifts to the right.

Nominal Exchange Rate Fluctuations

The Expected Future Exchange Rate

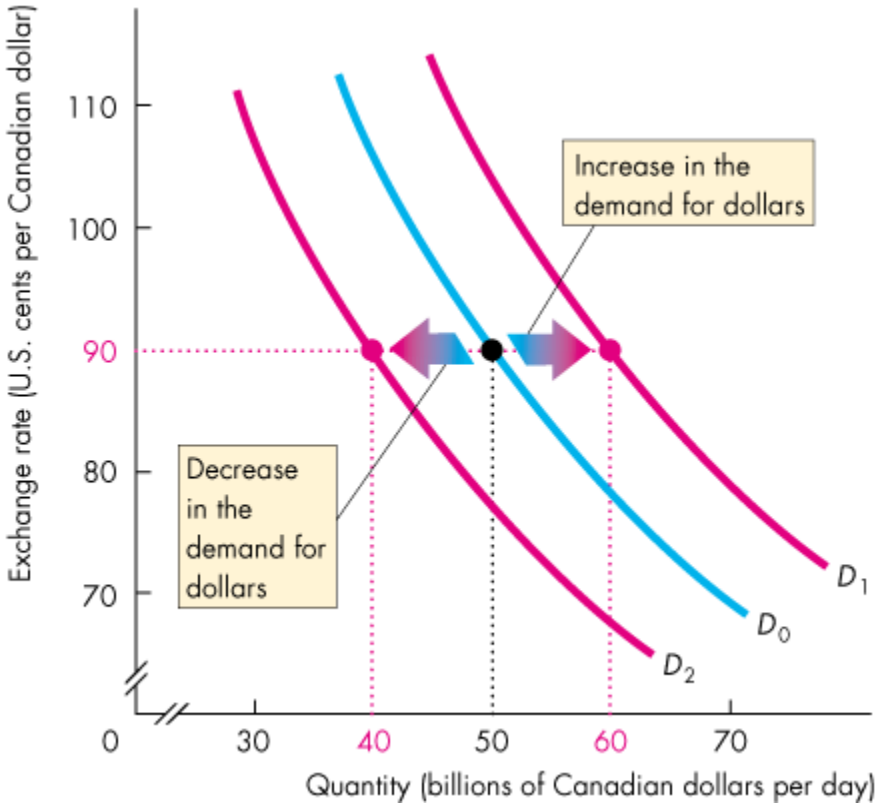
At a given current exchange rate, if the expected future exchange rate for domestic currency rises, the demand for domestic currency increases. The demand curve for domestic currency shifts rightward.

Nominal Exchange Rate Fluctuations

How the demand curve for Canadian dollars shifts in response to changes in

- Canadian exports
- The Canadian interest rate differential
- The expected future exchange rate

What are directions of shifts?



Nominal Exchange Rate Fluctuations

Changes in the Supply of domestic currency

A change in influencing factors on the quantity supplied of domestic currency, other than the exchange rate, brings a change in the supply of dollars.

These other influences include

- Domestic demand for imports
- Domestic interest rates relative to the foreign interest rate
- The expected future exchange rate

Nominal Exchange Rate Fluctuations

Domestic Demand for Imports

At a given exchange rate, if the domestic demand for imports increases, the supply of domestic currency in the foreign exchange market increases and the supply curve of domestic currency shifts rightward.

Domestic Interest Rate Relative to Foreign Interest Rate

If the domestic interest differential rises, the supply for domestic currency decreases and the supply curve of domestic currency shifts leftward.

Nominal Exchange Rate Fluctuations

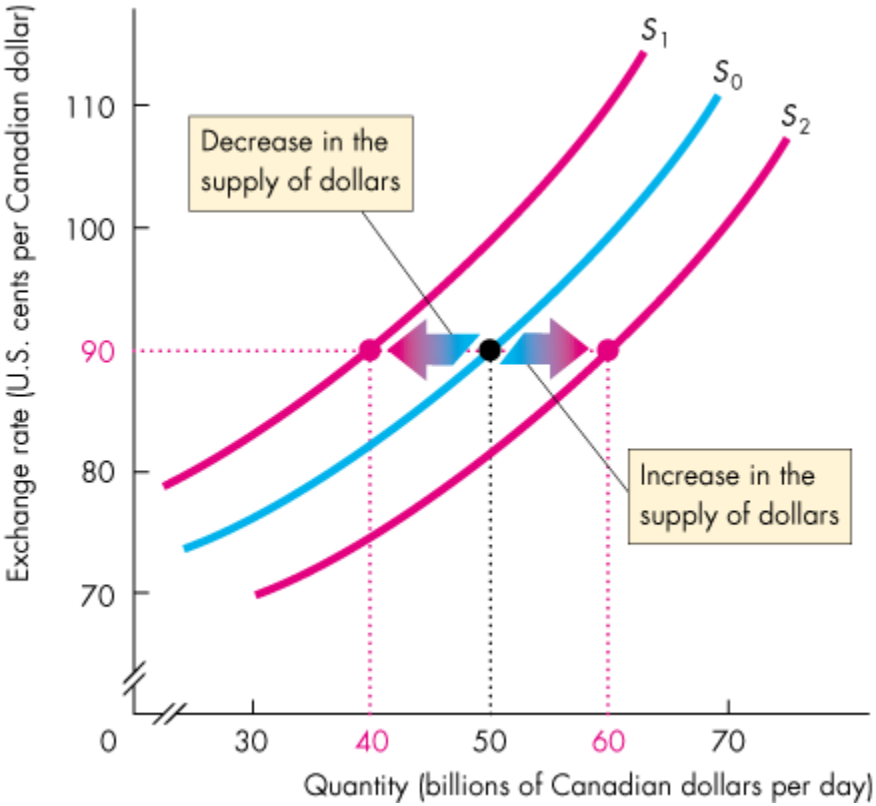
The Expected Future Exchange Rate

At a given current exchange rate, if the expected future exchange rate for domestic currency rises, the supply of domestic currency decreases and the supply curve of domestic curency shifts leftward.

Nominal Exchange Rate Fluctuations

How the supply curve of Canadian dollars shifts in response to changes in

- Canadian demand for imports
- Canadian interest rate differential
- Expected future exchange rate



Nominal Exchange Rate Fluctuations

Changes in the Exchange Rate

- If demand for Canadian dollars increases and supply does not change, the exchange rate rises.
- If demand for Canadian dollars decreases and supply does not change, the exchange rate falls.
- If supply of Canadian dollars increases and demand does not change, the exchange rate falls.
- If supply of Canadian dollars decreases and demand does not change, the exchange rate rises.

Arbitrage, Speculation, and Market Fundamentals

Arbitrage

Arbitrage is the practice of seeking to profit by buying in one market and selling for a higher price in another related market.

Arbitrage in the foreign exchange market and international loans and goods markets achieves four outcomes:

- The law of one price
- No round-trip profit
- Interest rate parity
- Purchasing power parity

Arbitrage, Speculation, and Market Fundamentals

The Law of One Price

The law of one price states that if an item can be traded in more than one place, the price will be the same in all locations.

No Round-Trip Profit

A round trip is using currency *A* to buy currency *B*, and then using *B* to buy *A*.

Arbitrage removes profit from all transactions of this type in equilibrium.

Arbitrage, Speculation, and Market Fundamentals

Interest Rate Parity

A currency is worth what it can earn.

The return on a currency is the interest rate on that currency plus the expected rate of appreciation over a given period.

When the rates of returns on two currencies are equal, we call it the **interest rate parity**.

Interest rate parity means *equal interest rates* when exchange rate changes are taken into account.

Market forces achieve interest rate parity very quickly.

Arbitrage, Speculation, and Market Fundamentals

Purchasing Power Parity

A currency is worth the value of goods and services that it will buy.

The quantity of goods and services that one unit of a currency buys differs from the quantity of goods and services that one unit of another currency buys.

When two quantities of currency can buy the same quantity of goods and services, we call the situation **purchasing power parity** (or PPP), which means *equal value of currency*.

Arbitrage, Speculation, and Market Fundamentals

Speculation

Speculation is trading on the *expectation* of making a profit.

Speculation contrast with arbitrage, which is trading on the certainty of making a profit.

Most foreign exchange transactions are based on speculation.

The expected future exchange rate influences both supply and demand, so it influences the current equilibrium exchange rate.

Arbitrage, Speculation, and Market Fundamentals

The Expected Future Exchange Rate

An expectation is a forecast.

Exchange rate forecasts are made over horizons that run from a few hours to months and years.

But exchange rate forecasts are hedged with a lot of uncertainty, there are many divergent forecasts, and the forecasts influence the outcome.

The dependence of today’s exchange rate on forecasts of tomorrow’s exchange rate can give rise to exchange rate volatility in the short run.

Arbitrage, Speculation, and Market Fundamentals

Exchange Rate Volatility

An exchange rate might rise one day and fall the next, as news about the influences on the exchange rate change the expected future exchange rate.

The influences of expectations and the constant arrival of news about the influences on supply and demand, make changes in the exchange rate impossible to predict.

But trends around which the exchange rate fluctuates are predictable and depend on market fundamentals.

Arbitrage, Speculation, and Market Fundamentals

The Real Exchange Rate

The **real exchange rate** is the relative price of goods and services produced in domestic economy to foreign-produced goods and services.

It measures the quantity of real GDP of other countries that a unit of domestic real GDP buys.

Real exchange rate (*RER*) is measured/calculated as:

$$RER = (E \times P)/P^*$$

where P is the domestic price level and P^* is the foreign price level,E is amount for foreign currency per unit of domestic currency.

Arbitrage, Speculation, and Market Fundamentals

If both countries produce identical goods, then the price levels expressed in the same currency would be the same and *RER* would equal 1.

In reality, countries produce different bundles of goods and the forces of demand and supply on goods markets determine P and P^* and the real exchange rate equals

$$RER = (E \times P)/P^*.$$

It turns out that, in the data, fluctuations in RER are mainly due to those in E.

Arbitrage, Speculation, and Market Fundamentals

Price Levels and Money

In the long run, *RER* is determined by the demand and supply in the markets for goods and services.

In the long run, *E* is determined by *RER* and the price levels. That is,

$$E = RER \times (P^*/P).$$

Thus in the long run, for a given real exchange rate, a change in the quantity of money brings a change in the price level *and* a change in nominal exchange rate.

Exchange Rate Policy

Three possible exchange rate policies are

- Flexible exchange rate
- Fixed exchange rate
- Crawling peg

Flexible Exchange Rate

A **flexible exchange rate** policy is one that permits the exchange rate to be determined by demand and supply with *no* direct intervention in the foreign exchange market by the central bank.

Exchange Rate Policy

Fixed Exchange Rate

A **fixed exchange rate** policy is one that pegs the exchange rate at a value decided by the government or central bank and is achieved by direct intervention in the foreign exchange market to block unregulated forces of demand and supply.

A fixed exchange rate requires active intervention in the foreign exchange market.

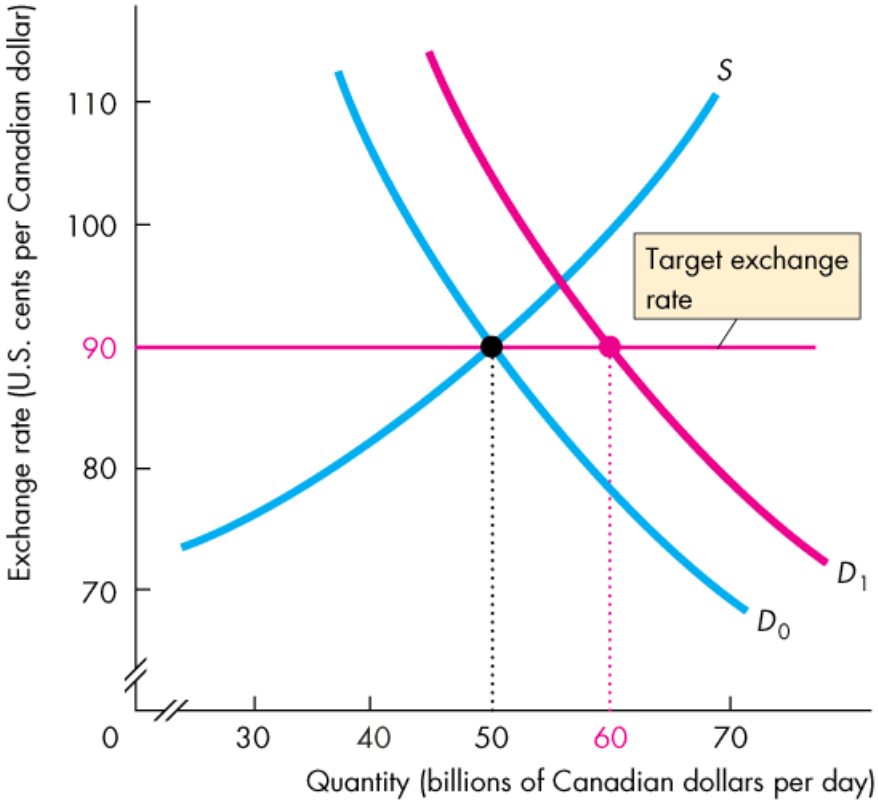
Exchange Rate Policy

How the central bank can intervene in the foreign exchange market.

Suppose that the target is 90 US cents per Canadian dollar.

If the demand for Canadian dollars increases, Bank of Canada sells Canadian dollars to increase supply.

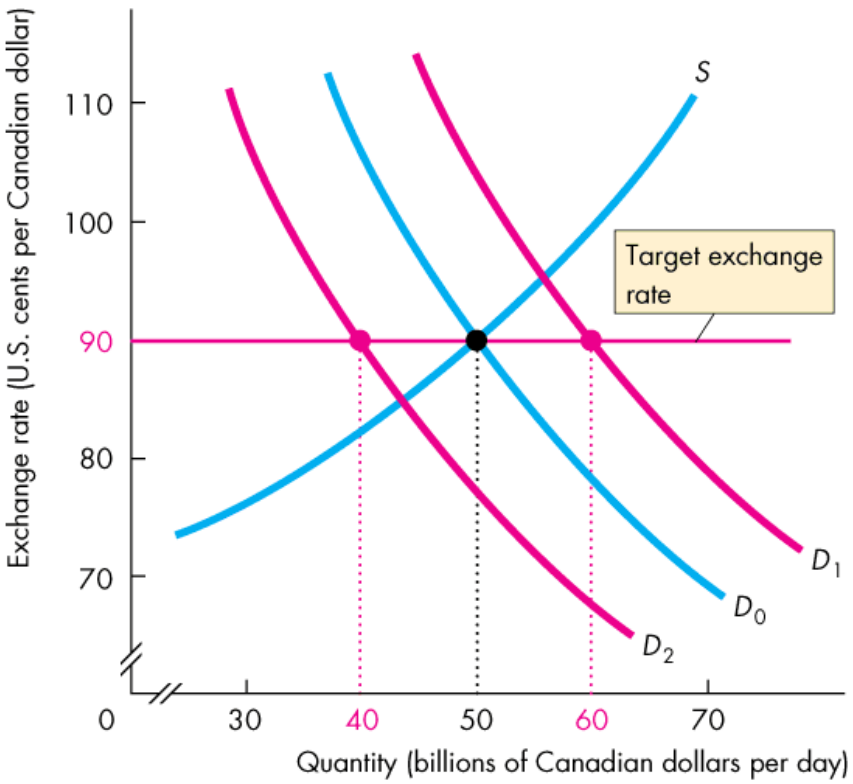
* But this effect is small.



Exchange Rate Policy

If demand for Canadian dollars decreases, the Bank of Canada buys Canadian dollars to decrease supply.

Persistent intervention on one side of the foreign exchange market cannot be sustained.



Exchange Rate Policy

Crawling Peg

A **crawling peg** is an exchange rate that follows a path determined by a decision of the government or the central bank and is achieved by active intervention in the market.

China is a country that operates a crawling peg.

A crawling peg works like a fixed exchange rate except that the target value changes.

Crawling peg is intended to avoid large swings in the exchange rate that might happen if expectations became volatile and to avoid the problem of running out of reserves, which can happen with a fixed exchange rate.

Financing International Trade

We’ve seen how the exchange rate is determined, but what is the effect of the exchange rate?

How does currency appreciation or depreciation influence Canadian international trade?

We record international transactions in the balance of payments accounts.

Financing International Trade

Balance of Payments Accounts

A country's **balance of payments accounts** records its international trading, borrowing, and lending.

There are three balance of payments accounts:

1. Current account
2. Capital and financial account
3. Official settlements account

Financing International Trade

The **current account** records

- receipts from exports of goods and services sold abroad,
- payments for imports of goods and services from abroad,
- net interest paid abroad, and
- net transfers (such as foreign aid payments).

The *current accounts balance* equals

exports − imports + net interest income + net transfers.

Financing International Trade

The **capital and financial account, or simply capital account,** records foreign investment in Canada minus Canadian investment abroad.

The **official settlements account** records the change in Canadian official reserves.

Canadian official reserves are the government's holdings of foreign currency.

If Canadian official reserves *increase*, the official settlements account is *negative*.

The sum of the balances of the three accounts always equals zero.

TABLE 9.1 Canadian Balance of Payments Accounts in 2016

Current account	**Billions of dollars**
Exports of goods and services	+629
Imports of goods and services	−677
Net interest income	−16
Net transfers	−3
Current account balance	−67
Capital and financial account	
Net foreign investment in Canada	+74
Statistical discrepancy	0
Capital and financial account balance	+74
Official settlements account	0
Offical settlements account balance	−7

Source of data: Statistics Canada, CANSIM Tables 376-0101 and 376-0104.

Financing International Trade

Borrowers and Lenders

A country that is borrowing more from the rest of the world than it is lending to it is called a **net borrower**.

A country that is lending more to the rest of the world than it is borrowing from it is called a **net lender**.

In 2009 and 2016, Canada was a net borrower, but between 1999 and 2009, Canada was a net lender.

Through most of the 1980s and 1990s, Canada was a net borrower from the rest of the world.

Financing International Trade

The Global Loanable Funds Market

The loanable funds market is global in an open economy.

Lenders want to earn the highest possible real interest rate and they will seek it by looking around the world.

Borrowers want to pay the lowest possible real interest rate and they will seek it by looking around the world.

Financial capital is mobile: It moves to the best advantage of lenders and borrowers.

Financing International Trade

Because lenders are free to seek the highest real interest rate and borrowers are free to seek the lowest real interest rate, the loanable funds market is a single, integrated, global market.

Funds flow into the country in which the real interest rate is highest and out of the country in which the real interest rate is lowest.

Financing International Trade

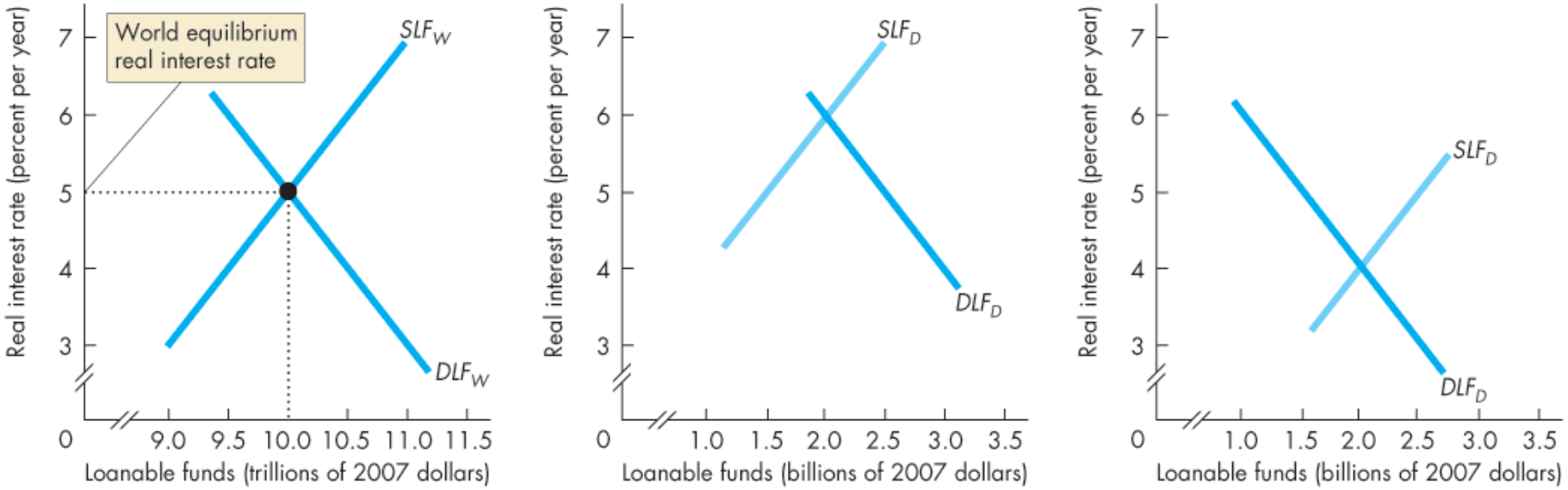
A country's loanable funds market connects with the global market through net exports.

If a country's net exports are *negative*, the rest of the world supplies funds to that country and the quantity of loanable funds in that country is greater than national saving.

If a country's net exports are *positive*, the country is a net supplier of funds to the rest of the world and the quantity of loanable funds in that country is less than national saving.

Financing International Trade

The world equilibrium real interest rate is 5 percent a year.

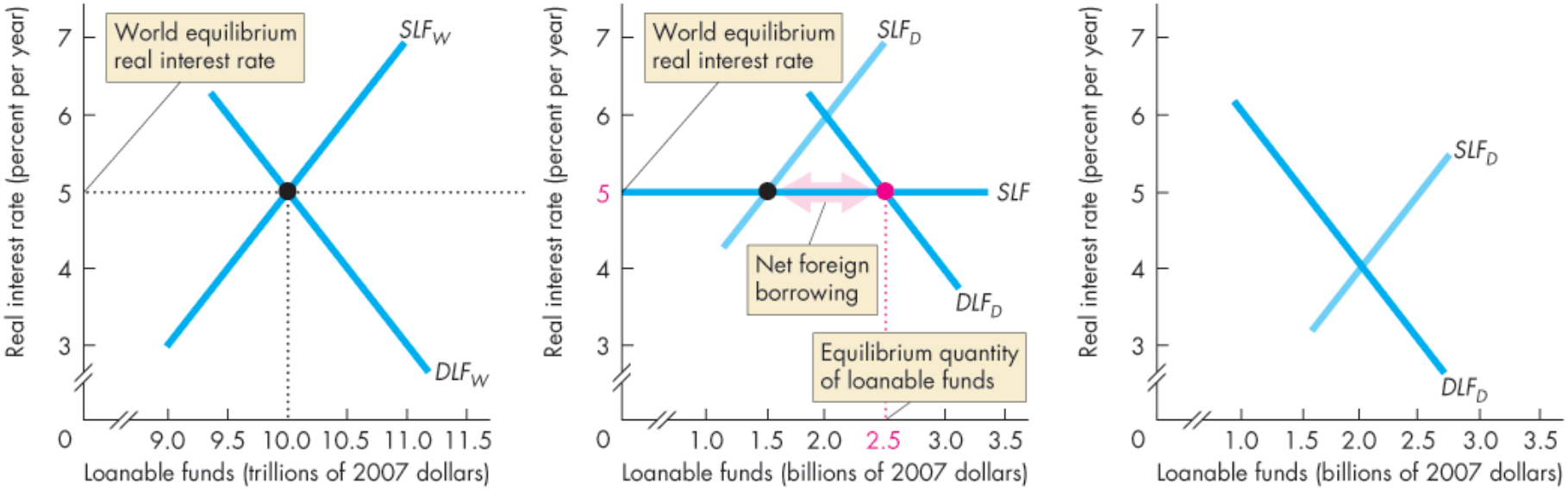


(a) The global market **(b) An international borrower** **(c) An international lender**

Financing International Trade

In part (b), at the world real interest rate, borrowers want more funds than the quantity supplied by domestic lenders.

The shortage of funds is made up by international borrowing.

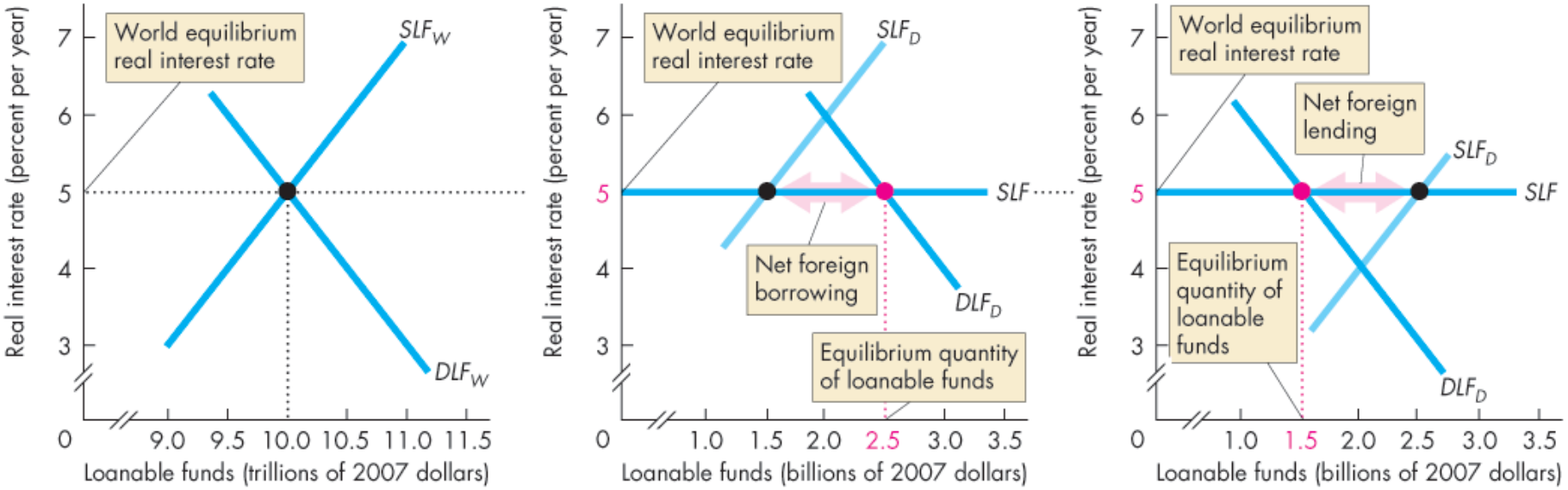


(a) The global market **(b) An international borrower** **(c) An international lender**

Financing International Trade

In part (c), at the world real interest rate, the quantity supplied by domestic lenders exceeds what domestic borrowers want.

The excess quantity supplied goes to foreign borrowers.



(a) The global market (b) An international borrower (c) An international lender

Financing International Trade

Current Account Balance

The current account balance (*CAB*) is

$$CAB = NX + \text{Net interest income} + \text{Net transfers}$$

The main item in the current account balance is net exports (*NX*).

The other two items are much smaller and don’t fluctuate much.

Financing International Trade

The **government sector surplus or deficit** is equal to net taxes, *T*, minus government expenditure on goods and services *G*.

The **private sector surplus or deficit** is saving, *S*, minus investment, *I*.

Net exports is equal to the sum of government sector balance and private sector balance:

$$NX = (T - G) + (S - I)$$

Financing International Trade

For Canada in 2016,

- Net exports were − $48 billion.
- Government sector balance was − $38 billion
- Private sector balance was − $10 billion

Net exports equals the sum of the government sector balance and the private sector balance.

TABLE 9.2 Net Exports, the Government Budget, Saving, and Investment

	Symbols and equations	Canada in 2016 (billions of dollars)
(a) Variables		
Exports*	X	629
Imports*	M	677
Government expenditure	G	509
Net taxes	T	471
Investment	I	385
Saving	S	375
(b) Balances		
Net exports	$X - M$	$629 - 677 = -48$
Government sector	$T - G$	$471 - 509 = -38$
Private sector	$S - I$	$375 - 385 = -10$
(c) Relationship among balances		
National accounts	$Y = C + I + G + X - M$ $= C + S + T$	
Rearranging:	$X - M = S - I + T - G$	
Net exports equals:	$X - M$	−48
Government sector plus	$T - G$	−38
Private sector	$S - I$	−10

Source of data: Statistics Canada, CANSIM Tables 380-0002 and 380-0064.

* The national income and expenditure accounts measures of exports and imports are slightly different from the balance of payments accounts measures in Table 9.1 on p. 228.

Financing International Trade

Where is the Exchange Rate?

In the short run, a fall in the nominal exchange rate lowers the real exchange rate, which makes our imports more costly and our exports more competitive.

So in the short run, fall in the nominal exchange rate decreases the current account deficit.

But in the long run, a change in the nominal exchange rate leaves the real exchange rate unchanged.

So in the long run, the nominal exchange rate plays no role in influencing the current account balance.